



Your Private Backpack Coach™

COACH LIBRARY · 2 OF 3

Any pattern. Any size. Any yarn.

The Resize & Yarn-Swap Guide: how to make any of your 50 backpacks for any age, and how to swap one yarn for another without ruining the bag.

- ✓ Size chart by age, from toddler to adult
- ✓ The stitch-count formula with worked examples
- ✓ Yarn, hook and quantity swaps — with the math done

Before you change anything

Every resize and every yarn swap starts in the same place: a 10-minute gauge swatch. It tells you how many stitches and rounds *your* hands make per centimeter with *your* yarn — the only number that matters.

The gauge swatch in 6 steps

1. With the yarn and hook you'll use for the bag, ch 17 (for t-shirt yarn: ch 13).
2. Work in the pattern's main stitch (usually sc) until the piece is a square of about 15 cm / 6 in.
3. Don't fasten off yet. Lay it flat without stretching it.
4. Place 2 pins exactly 10 cm / 4 in apart, horizontally, in the center. Count the stitches between them: that's **S**.
5. Do the same vertically and count the rows/rounds: that's **R**.
6. Write it down: **stitches per cm** = $S \div 10$ and **row height (cm)** = $10 \div R$.

More stitches than the pattern? → go up one hook size.

Fewer stitches than the pattern? → go down one hook size.

Repeat until your swatch matches the gauge on the pattern's spec sheet (page 1 of each pattern).

WHY IT MATTERS

1 stitch of difference in 10 cm on a 64-stitch bag changes its width by about 4 cm / 1.5 in. That's the difference between "fits a lunchbox" and "doesn't close".

The size chart

Finished measurements of the bag, lying flat. Straps are measured end to end, each one, before sewing (the same way the patterns give them). Use this as your target, then apply the formula on the next page.

SIZE	HEIGHT	WIDTH	DEPTH (BASE)	FLAP DROP	STRAP (EACH)	ROUND BASE Ø
Toddler · 2–3 yrs	22 cm / 8.75 in	19 cm / 7.5 in	7 cm / 2.75 in	10 cm / 4 in	32 cm / 12.5 in	14 cm / 5.5 in
Preschool · 4–5 yrs	26 cm / 10.25 in	21 cm / 8.25 in	8 cm / 3.25 in	12 cm / 4.75 in	38 cm / 15 in	16 cm / 6.25 in
School · 6–8 yrs	30 cm / 11.75 in	24 cm / 9.5 in	9 cm / 3.5 in	13 cm / 5 in	45 cm / 17.75 in	18 cm / 7 in
Big kid · 9–12 yrs	34 cm / 13.5 in	27 cm / 10.75 in	10 cm / 4 in	15 cm / 6 in	52 cm / 20.5 in	20 cm / 7.75 in
Adult mini	24 cm / 9.5 in	21 cm / 8.25 in	9 cm / 3.5 in	12 cm / 4.75 in	55 cm / 21.75 in	18 cm / 7 in
Adult standard	36 cm / 14.25 in	29 cm / 11.5 in	12 cm / 4.75 in	16 cm / 6.25 in	60 cm / 23.5 in	23 cm / 9 in

MEASURE THE CHILD, NOT THE AGE

Kids of the same age vary a lot. Two checks: **1)** the bag height should be the distance from the top of the child's shoulder to the waist, minus 5 cm / 2 in. **2)** Hang a string from the top of the back over the shoulder to the hip: that's the strap length.

The resize formula

Two ways to resize. Method A works with the pattern you have. Method B builds the count from your swatch. Both take 5 minutes with a calculator.

Method A • The scale factor

factor = target width ÷ pattern width

new stitch count = pattern count × factor

...then round to the nearest multiple of the pattern repeat (the number of stitches in the brackets, e.g. "(inc, 6 sc)" = repeat of 8).

WORKED EXAMPLE

A pattern for 6–8 yrs has **64 stitches around** the body (repeat of 8) and measures **26.5 cm / 10.5 in** wide flat. You want it for a 4-year-old: **23 cm / 9 in**.

factor = $23 \div 26.5 = 0.868$ · new count = $64 \times 0.868 = 55.5$ → nearest multiple of 8 = **56 stitches**.

The base is a flat circle that grows 8 sts per round (8, 16, 24... 64 = 8 rounds). For 56 stitches you stop at **round 7**, then work the walls on 56.

Height: count rounds, not the pattern

rounds needed = target height ÷ your row height

Example: target height 26 cm / 10.25 in; your swatch gives 12.5 rounds per 10 cm → row height = 0.8 cm. $26 \div 0.8 = 32.5$ → **33 rounds** from the base to the top. If the pattern has a top band (eyelets + 2 rounds), count those inside the 33: work 30 body rounds + 3 band rounds.

Method B • Build the count from your swatch

PIECE	FORMULA	EXAMPLE (1.2 STS/CM, ROW HEIGHT 0.8 CM)
Round base	Start 6 sc in MR (8 if hdc, 12 if dc), add the same number every round. Rounds = $(\varnothing \div 2) \div$ row height.	$\varnothing 16 \text{ cm} \rightarrow 8 \div 0.8 = 10 \text{ rounds}$ of +6 → 60 sts
Oval base	Foundation ch = (base length – base width) × sts per cm, + 1. Rounds = $(\text{base width} \div 2) \div$ row height.	$21 \times 8 \text{ cm} \rightarrow (21 - 8) \times 1.2 = 15.6 \rightarrow 16 + 1 = \text{ch } 17$; $4 \div 0.8 = 5 \text{ rounds}$
Flat panel / flap	Starting ch = width × sts per cm (+1 turning ch for sc).	Flap 20 cm → $20 \times 1.2 = 24 \text{ sts}$, ch 25

PIECE	FORMULA	EXAMPLE (1.2 STS/CM, ROW HEIGHT 0.8 CM)
Straps	Length in rows = strap length ÷ row height. Keep strap width: 3 cm / 1.25 in kids, 4 cm / 1.5 in adults.	38 cm ÷ 0.8 = 48 rows of 4 sc

Method C • No math (for small changes)

- One hook size up (e.g. 5 mm → 5.5 mm) ≈ **8–12% bigger** in every direction. One size down ≈ 8–12% smaller.
- Same pattern in t-shirt yarn instead of bulky cotton ≈ **30–50% bigger** (and heavier).
- Two hook sizes is the limit: beyond that the fabric gets too loose (floppy bag) or too stiff (hurts your hands).

WHAT NEVER SCALES

Hardware and details keep their size: a 2.5 cm / 1 in clasp, eyelets every 3–4 stitches, a 3 cm / 1.25 in strap for kids. Resize the body, base, flap and strap length only. Animal faces and appliqués: make them at the original size for kids 4+, and one hook size smaller for toddler bags.

The yarn-swap guide

You can make any pattern in almost any yarn — as long as you respect 4 rules. Break one and the bag comes out floppy, tiny or twice as heavy.

RULE 1 · SAME WEIGHT CLASS

Swap within the same thickness (see the table). A different thickness means resizing (Method B).

RULE 2 · RESPECT THE FIBER

Cotton holds shape and doesn't stretch; acrylic stretches; t-shirt yarn stretches and is heavy; cord is rigid.

RULE 3 · MATCH THE GAUGE

Always swatch the new yarn and adjust the hook until it matches the pattern's gauge.

RULE 4 · BUY BY LENGTH

Compare meters/yards, never grams or number of balls. 200 g of t-shirt yarn and 200 g of cotton are very different lengths.

Yarn table for backpacks

YARN	CYC WEIGHT	LABEL HOOK	HOOK FOR BAGS	SC / 10 CM (4 IN)	ROWS / 10 CM	HOW IT BEHAVES
Cotton DK / light	3	4 mm (G-6)	3.5 mm (E-4)	19–21	21–23	Firm, crisp stitches, no stretch. Needs a lining for adult sizes.
Worsted cotton or cotton blend	4	5 mm (H-8)	4–4.5 mm (G-6 / 7)	15–17	17–19	The yarn used in most of the collection. Structured and washable, a bit heavy.
Worsted acrylic	4	5 mm (H-8)	4 mm (G-6)	15–17	17–19	Light and cheap, but stretches: line the straps.
Bulky cotton / chunky	5	6.5 mm (K-10.5)	5.5–6 mm (I-9 / J-10)	11–13	12–14	Fast, soft and cozy. Firm only with a smaller hook.
Cotton cord / macramé cord 3–5 mm	5	—	5–6 mm (H-8 / J-10)	11–14	12–15	Very structured, great stitch definition, zero stretch. Ideal for adult bags.
T-shirt yarn, fine (strip ~5–7 mm)	6	8 mm (L-11)	7 mm (K-10½ / 7 mm)	10–11	10–12	Used in several patterns of the collection. Stretchy, chunky, forgiving.

YARN	CYC WEIGHT	LABEL HOOK	HOOK FOR BAGS	SC / 10 CM (4 IN)	ROWS / 10 CM	HOW IT BEHAVES
T-shirt yarn, standard (strip ~1 cm)	6–7	9–10 mm (M/N-13 / N/P-15)	7–8 mm (7 mm / L-11)	7–9	7–9	Super fast, very firm. Hard on the hands — take breaks.
Jute / raffia	4–5	5–6 mm (H- 8 / J-10)	4.5–5 mm (7 / H-8)	13–16	14–16	Stiff and rustic, holds any shape. Scratchy on bare shoulders: line the straps.

Typical values for firm bag fabric in sc. Every brand and every pair of hands is different — the swatch always wins.

The #1 question: can I swap t-shirt yarn for cotton?

Yes — in three ways, from easiest to most work:

OPTION	HOW	RESULT
1 • Cotton cord 4–5 mm	Swap directly, hook 6–7 mm (J-10 / K-10½). Swatch to confirm.	Closest gauge to fine t-shirt yarn. More structured, lighter.
2 • Hold strands together	2 strands of worsted cotton held together ≈ bulky; 3 strands ≈ fine t-shirt yarn. Hook 6–7 mm.	Same pattern, same counts. Uses 2–3× the yarn length.
3 • Single worsted cotton	Use Method B with your swatch: counts roughly double (e.g. 60 sts → 110–120 sts).	Finer, more elegant bag. Takes 2–3× longer. Add a lining.

THE OTHER WAY ROUND: COTTON PATTERN IN T-SHIRT YARN

Fine t-shirt yarn with a 5–6 mm hook ≈ bulky cotton. For a pattern written in worsted, use Method B: counts go down roughly by half (e.g. 96 sts → 48–54 sts), and the bag will be much heavier — perfect for a sturdy school bag.

How much yarn do I need?

Two methods. The first uses the pattern's quantity. The second works for any design and any yarn — even your own.

Method 1 • Convert the pattern's quantity

meters needed = pattern grams × (meters ÷ grams on the pattern yarn's label)

balls to buy = meters needed ÷ meters per ball of your yarn × 1.15

Example: the pattern uses 300 g of a yarn labeled 100 g = 170 m → $300 \times 1.7 = \mathbf{510\ m}$. Your cotton is 50 g = 85 m per ball → $510 \div 85 \times 1.15 = 6.9 \rightarrow \mathbf{\text{buy 7 balls}}$. (1 yd = 0.91 m; 1 m = 1.09 yd.)

Method 2 • Weigh your swatch (works for anything)

1. Make a 10 × 10 cm / 4 × 4 in square in the bag's stitch and weigh it on a kitchen scale: that's **g** (grams per 100 cm²).
2. Calculate the bag's area: **$2 \times (H \times W) + 2 \times (H \times D) + (W \times D) + (W \times \text{flap drop}) + 2 \times (\text{strap length} \times \text{strap width})$** .
3. grams needed = area ÷ 100 × g × **1.15** (15% extra for joins, ends and mistakes).

WORKED EXAMPLE • PRESCHOOL SIZE (4–5 YRS)

H 26, W 21, D 8, flap 12, straps 2 × 38 × 3 cm → area = 1,092 + 416 + 168 + 252 + 228 = **2,156 cm²**.

Fine t-shirt yarn, swatch 14 g → $21.56 \times 14 = 302\ \text{g} \times 1.15 = \mathbf{347\ g}$ → one 500 g cone.

Worsted cotton, swatch 4 g → $21.56 \times 4 = 86\ \text{g} \times 1.15 = \mathbf{99\ g}$ → 3 balls of 50 g (2 balls = 100 g is not enough).

BUY IT ALL AT ONCE

Buy the total plus one extra ball/cone **from the same dye lot** (the lot number is on the label). T-shirt yarn changes color and thickness between cones — see Fix 36 in The Fix-It Library.

Quick swap cheat sheet

YOU HAVE / WANT	DO THIS
Pattern in t-shirt yarn, you have cotton cord	Swap directly, hook 6 mm (J-10), swatch. Buy by length.
Pattern in t-shirt yarn, you have worsted cotton	Hold 3 strands together with a 6–7 mm hook, or use Method B.
Pattern in cotton, you want acrylic	Same weight, same hook. Line the straps (acrylic stretches).
You want a softer bag for a toddler	Bulky cotton instead of t-shirt yarn, Method B for the counts. Lighter on small shoulders.
You want a stiffer adult bag	Cotton cord or jute, one hook size smaller than the label, plus a lining with interfacing.
The bag must be washable	Cotton or cotton cord. Avoid jute. T-shirt yarn: hand wash cold only.
You're selling the bag	Pick a yarn you can buy again in the same color for months — buyers order the same bag in another size.
Color work (animal faces, stripes)	Keep all colors the same yarn and weight. Mixing cotton with t-shirt yarn in one piece warps the fabric.

Hook sizes: mm ↔ US

METRIC	US	METRIC	US
3.5 mm	E-4	6.5 mm	K-10.5
3.75 mm	F-5	7 mm	— (sold as 7 mm)
4 mm	G-6	8 mm	L-11
4.5 mm	7	9 mm	M/N-13
5 mm	H-8	10 mm	N/P-15
5.5 mm	I-9	12 mm	— (sold as 12 mm)
6 mm	J-10	15 mm	P/Q

Always go by the mm size — US letters vary slightly between brands.

RESIZE OR SWAP IN 30 SECONDS

Open Your Private Backpack Coach™ and write: "*[Pattern name], I want it for a [age] with [yarn], my swatch is [S] stitches and [R] rows in 10 cm.*" The coach does the math for base, body, flap and straps.

Every pattern. Every size.

With a swatch, a calculator and this guide, the 50 patterns in your collection become 300 — every age, every yarn, every color you want.